



ಬಿ.ಎಂ.ಎಸ್. ತಾಂತ್ರಿಕ ಮತ್ತು ವ್ಯವಸ್ಥಾಪನಾ ಮಹಾವಿದ್ಯಾಲಯ

**BMS INSTITUTE OF TECHNOLOGY & MANAGEMENT**  
(Autonomous Institution Under VTU)

Yelahanka, Bengaluru -560119



**A Major-project Phase 1 Report**

*on*

**”Real-Time AI/ML based Smishing and Quishing Detection and Prevention System”**

*Submitted in partial fulfillment for the award of*

**BACHELOR OF ENGINEERING**

*in*

**COMPUTER SCIENCE AND ENGINEERING**

*by*

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**2025 – 2026**



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**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**



**CERTIFICATE**

This is to certify that the Major-Project work (BCS606) entitled **”Real-Time AI/ML based Smishing and Quishing Detection and Prevention System”** is a bonafide work carried out by **Mr. S MD Umar Talha Azeez(1BY23CS190), Mr. Nithin Naik R (1BY23CS152),Mr. Prashant Kushwaha(1BY23CS161) ,Mr. Sagar S H(1BY24CS415)** in partial fulfillment for the award of **Bachelor of Engineering Degree in Computer Science and Engineering** of the Visvesvaraya Technological University, Belagavi during the year 2025–26. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in this report. The Major -project phase 1 report has been approved as it satisfies the academic requirements with respect to Major-project work for the B.E Degree.

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## **ABSTRACT**

Quishing and smishing are increasingly common phishing techniques that target users through QR codes and SMS messages. These attacks exploit user trust, particularly in online payments, banking interactions, digital communication, and service verification workflows. As QR codes and text messages become part of everyday digital activity, attackers use them to deliver malicious URLs, social engineering prompts, and deceptive instructions that can lead to credential theft, fraud, or device compromise.

The proposed system, Real Time AI/ML Based Quishing and Smishing Detection and Prevention System, is designed to identify such threats in real time using a unified mobile-based security framework. It continuously monitors incoming SMS messages and scanned QR codes, extracts relevant textual and URL-based features, and analyzes them using artificial intelligence and machine learning methods. The SMS stream is handled through natural language processing techniques that help identify urgency cues, suspicious sender behavior, and phishing-oriented language patterns. QR-based inputs are decoded and analyzed to inspect URL structure, domain details, special characters, and other indicators of suspicious behavior.

The system introduces a hybrid detection pipeline in which smishing and quishing are handled separately but correlated through a cross-verification module. This allows the platform to detect contextual phishing attacks, such as an SMS message that appears to come from a financial institution but leads to a non-banking or unrelated domain. Based on this analysis, the system assigns a risk score and provides preventive actions such as user alerts, link blocking, and safety recommendations.

The proposed solution emphasizes timely detection, improved accuracy, low manual effort, and better user awareness. It is designed as a practical mobile security product suitable for mobile banking users, students, elderly users, online shoppers, and organizations that rely on QR- and SMS-based communication. By combining detection and prevention in a single framework, the system aims to reduce financial fraud, identity theft, and other forms of digital exploitation.

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## Declaration

We, hereby declare that the major-project (BCS606) titled “**Real-Time AI/ML based Smishing and Quishing Detection and Prevention System**” is a record of original major-project work undertaken for partial fulfillment of Bachelor of Engineering in Computer Science and Engineering of the Visvesvaraya Technological University, Belagavi during the year 2025–26. We have completed this major-project work under the guidance of **Prof. Brunda S, Assistant Professor**.

I also declare that this major-project report has not been submitted for the award of any degree, diploma, associateship, fellowship or other titles anywhere else.

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## **INTRODUCTION**

In recent years, the rapid growth of smartphones, digital payment platforms, online banking, e-commerce applications, and QR-code-based services has significantly increased the use of mobile devices for everyday activities. Along with technological advancements, cyber threats have also evolved rapidly, making phishing attacks one of the most dangerous cybersecurity challenges in the digital world. Attackers continuously exploit user trust and human behavior to steal sensitive information such as passwords, banking credentials, OTPs, and personal data.

Traditional phishing attacks mainly targeted users through fake emails and fraudulent websites. However, modern attackers increasingly use mobile communication channels such as SMS messages and QR codes to perform phishing attacks. These attacks are commonly known as Smishing (SMS phishing) and Quishing (QR-code phishing). Smishing attacks use fraudulent SMS messages containing malicious links or deceptive information to trick users into revealing confidential data. Similarly, quishing attacks use malicious QR codes that redirect users to fake websites, payment portals, or malware-infected pages without their knowledge.

The widespread adoption of QR-code-based payments and mobile banking applications has increased the success rate of quishing attacks. Users often scan QR codes without verifying their authenticity because QR codes visually appear safe and trustworthy. Similarly, phishing SMS messages create urgency, fear, rewards, or fake notifications to manipulate users emotionally and force them to click malicious links. These attacks may result in financial fraud, identity theft, privacy breaches, and unauthorized access to personal accounts.

Existing phishing detection systems mainly focus on email phishing and traditional web-based attacks. Many current systems rely on blacklist mechanisms or static rule-based approaches that are ineffective against modern phishing techniques. Attackers continuously generate new domains, modify URLs, and change phishing strategies to bypass traditional security systems. In addition, many existing solutions analyze SMS messages and QR codes separately, which reduces the effectiveness of detecting combined or context-based phishing attacks.

Overall, the proposed system offers an intelligent, scalable, and efficient solution for addressing modern smishing and quishing attacks in mobile environments. It demonstrates how AI and ML technologies can be effectively utilized to build real-time cybersecurity systems capable of adapting to evolving phishing threats and protecting users from digital fraud.

## **PROBLEM STATEMENT**

In the contemporary mobile computing ecosystem, users rely heavily on Quick Response (QR) codes and Short Message Service (SMS) channels for critical operations, including mobile banking, point-of-sale payments, multi-factor authentication (MFA), and official communications. While these technologies offer unparalleled convenience, their ubiquity has exponentially expanded the mobile attack surface. Specifically, Quishing (QR-code-based phishing) and Smishing (SMS-based phishing) have emerged as highly effective vectors for social engineering. Attackers exploit structural user trust by delivering malicious payloads, credential-harvesting links, and deceptive prompts directly to personal devices, systematically bypassing established perimeter defenses.

Conventional mobile security frameworks fail to address these vectors because they rely heavily on centralized blacklists, static signature matching, and remote reputation-lookup APIs. This reactive paradigm is structurally incapable of mitigating modern zero-day phishing attacks. Attackers routinely circumvent these defenses by utilizing programmatic techniques such as dynamic domain generation algorithms (DGAs), automated URL shortening services, and episodic, short-lived hosting infrastructure. Because traditional systems require an initial victim to report a threat before updating a centralized database, a critical window of vulnerability exists during which newly minted malicious domains remain undetected.

The most sophisticated security failure, however, stems from the siloed analysis of multi-vector threats. Current mobile security tools analyze SMS text strings and QR code URLs as isolated, mutually exclusive data streams. Attackers deliberately exploit this structural blind spot through contextual deception, where a malicious SMS message may flawlessly replicate the brand identity, syntax, and context of a legitimate financial institution or courier service, passing local text filters, while the embedded URL or QR code points to an unfamiliar, adversarial domain. Because traditional systems lack the correlation logic required to map the claimed semantic identity of the text against the actual structural destination of the link, cross-vector brand impersonation and typosquatting attacks seamlessly evade detection.

## OBJECTIVES

The primary aim of this research is to engineer a sophisticated mobile security architecture capable of the real-time identification of quishing (QR-code phishing) and smishing (SMS phishing) attacks, thereby preempting user interaction with malicious payloads. The following objectives are strategically partitioned across detection, analysis, prevention, and cognitive security to ensure the final system provides active protection at the point of risk:

- To implement a persistent background service for the autonomous surveillance of incoming SMS traffic and scanned/embedded QR codes, ensuring that anomalous communication vectors are intercepted at the immediate point of ingestion.
- To develop a robust preprocessing pipeline utilizing Natural Language Processing (NLP) for SMS telemetry and Computer Vision (CV) for QR imagery to facilitate high-fidelity feature extraction and architectural consistency during downstream analysis.
- To deploy advanced Machine Learning models specialized in linguistic and contextual analysis to identify smishing attempts based on indicators of urgency, sender behavioral anomalies, and adversarial language patterns.
- To engineer a dual-layered detection mechanism for QR codes that combines visual integrity analysis with URL safety classification models to isolate unsafe redirects, domain spoofing, and unconventional payloads.

### 3.1 Functional Objectives

The functional requirements of the system are categorized into monitoring, decoding, classification, scoring, alerting, and prevention. These goals ensure that the technical pipeline translates into a seamless end-user experience:

- Ensures the automated capture of data streams without manual intervention and the high-accuracy transformation of graphical QR data into machine-readable URLs or strings.
- Provides the analytical core where content is triaged (Safe, Suspicious, or Malicious) and converted into a standardized risk index for intuitive user interpretation.
- Provides the analytical core where content is triaged (Safe, Suspicious, or Malicious) and converted into a standardized risk index for intuitive user interpretation.

By fulfilling these objectives, the system bridges the gap between complex machine learning pipelines and daily mobile utility.

## LITERATURE SURVEY

Phishing attacks have evolved rapidly with the increasing use of smartphones, online banking, QR-code payments, digital wallets, and instant messaging services. Traditional phishing attacks mainly targeted users through emails and fake websites, but modern attackers now exploit mobile communication channels such as SMS messages and QR codes. These attack methods are commonly known as Smishing (SMS phishing) and Quishing (QR-code phishing). Because users frequently trust text messages and scan QR codes without proper verification, these attacks have become highly effective and dangerous.

Several research works have focused on the use of Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL) techniques for phishing detection. Existing systems commonly analyze suspicious URLs, website structures, message contents, sender information, and user behavior patterns to classify communications as legitimate or malicious. Machine learning algorithms such as Random Forest, Support Vector Machine (SVM), Decision Tree, Naive Bayes, XGBoost, and Logistic Regression have been widely used for phishing detection because of their ability to identify hidden patterns from large datasets.

Recent studies have also explored deep learning approaches such as Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), and transformer-based models like TinyBERT. These models improve phishing detection by understanding the context and semantics of messages instead of relying only on fixed rules or keywords. Deep learning methods are especially useful in identifying smishing attacks because phishing messages often use emotional manipulation, urgency, fake rewards, or threatening language to deceive users.

Research on QR-code security has gained importance due to the rapid growth of QR-based payments, ticketing systems, advertisements, and authentication mechanisms. QR codes can easily hide malicious URLs because users cannot visually identify the actual destination before scanning. Existing research has proposed QR-code analysis systems that decode QR payloads and inspect URL characteristics such as domain length, suspicious characters, redirects, encoded strings, and unsafe domains. Some studies also combine machine learning and deep learning methods to classify QR-code-related URLs as safe or malicious.

Many existing phishing detection systems achieve good classification accuracy, but they still face several limitations. Most systems focus on a single attack vector such as website phishing, email phishing, SMS phishing, or QR-code phishing independently. Very few systems provide a unified framework capable of handling both smishing and quishing attacks together. In many real-world attacks, phishing messages and QR codes are used together to increase user trust and bypass traditional security systems. Therefore, separate analysis of SMS messages and QR codes may fail to detect coordinated or contextual phishing attacks.

Another limitation identified in the literature is the dependency on blacklist-based detection methods. Blacklists can identify only previously known malicious websites or URLs and are ineffective against newly generated phishing domains or zero-day attacks. Attackers continuously modify URLs, use URL shorteners, generate fake domains, and imitate trusted brands, making static detection approaches insufficient. AI and ML-based methods improve adaptability, but many existing systems still require large datasets, continuous retraining, and high computational resources.

The literature also highlights the challenge of implementing real-time phishing detection on mobile devices. Since smartphones have limited processing power and battery constraints compared to desktop systems, lightweight and efficient AI models are necessary. Researchers have proposed optimized models suitable for Android devices, but many solutions still focus only on detection and do not provide complete prevention mechanisms such as blocking malicious links, generating alerts, or warning users before interaction.

Another important observation from the literature is the lack of contextual verification. Existing systems usually analyze either the SMS content or the URL independently. However, modern phishing attacks often use social engineering strategies where an SMS message appears to originate from a trusted source, while the QR code or URL redirects users to a completely unrelated or malicious website. Without comparing the message context and the destination URL together, phishing detection accuracy decreases.

The literature survey clearly indicates that phishing attacks are becoming more intelligent, mobile-oriented, and difficult to detect using traditional security techniques. Although significant progress has been made using AI, ML, NLP, and deep learning methods, there is still a need for a unified, lightweight, real-time mobile security framework that can analyze SMS messages, QR codes, URLs, and contextual relationships together.

To overcome these limitations, the proposed project introduces a Real Time AI/ML Based Quishing and Smishing Detection and Prevention System. The system combines SMS monitoring, QR-code analysis, URL verification, AI/ML-based classification, contextual cross-verification, risk scoring, and real-time prevention mechanisms into a single integrated framework. This approach aims to improve detection accuracy, reduce false positives, provide faster alerts, and protect users from modern phishing attacks in real-time mobile environments.

| Paper, Author, Year                                                                                               | Purpose                                                                                                                                                          | Algorithms                                  | Limitations                                                                                                                                       | Results                                                                                                                             |
|-------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Abdulrehman Arif et al.,2025, AI-Driven Edge Computing for IoT: Revolutionizing Phishing Detection and Mitigation | To integrate AI with edge computing for real-time phishing detection in IoT environments                                                                         | DNN, Federated Learning, RL, Lightweight AI | Scalability issues, privacy concerns, resource constraints in IoT                                                                                 | Achieved 95 detection accuracy with low latency and real-time processing                                                            |
| Daniela Popescu and Laura Radu ,2025,AI in Phishing Detection: A Bibliometric Review                              | To analyze research trends, algorithms, and evolution of AI-based phishing detection                                                                             | NLP, Hybrid Models                          | High false positives, need for large datasets, lack of explainability                                                                             | Identified shift from ML → DL with improved accuracy and hybrid approaches                                                          |
| Neha Yogesh Nagdeve et al.,2025,AI/ML-Based Phishing Detection and Prevention System                              | To develop an intelligent phishing detection system using AI/ML that can analyze emails, URLs, and websites to identify phishing attacks and prevent data theft. | Random Forest, Decision Tree                | Requires large datasets, high computational resources, and may face challenges in adapting to completely new phishing patterns (zero-day attacks) | Improved real-time detection accuracy and adaptability using AI techniques; better performance than traditional rule-based systems. |
| Asadullah Safi and Satwinder Singh,2024, A Systematic Literature Review on Phishing Website Detection Techniques. | To analyze and compare different phishing detection techniques (ML, DL, heuristic, list-based) and identify the most effective approaches and datasets.          | Random Forest, SVM, CNN                     | Mostly focused on website phishing; lacks real-time implementation aspects and limited discussion on mobile-based phishing (SMS/QR).              | Machine Learning and Deep Learning models (especially CNN) achieved highest accuracy ( 99); ML is the most widely used approach.    |

|                                                                                                                                                                                                          |                                                                                                                                                                   |                         |                                                                                                                                                                            |                                                                                                                                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bello Bello Musa, Adamu Ahmad Bahago, Nasir Auwal Muhammad, Dr. Fakhrun Jamal, 2025, The Impact of Emerging Phishing Threats: Assessing Quishing and LLM-generated Phishing Emails against Organizations | To study how quishing and LLM-generated phishing emails affect organizations, and to check how effective current defenses are against these new phishing threats. | Random Forest, SVM, CNN | It is mainly a review paper, so it does not present a new experimental model, dataset, or benchmark test. It also does not give a fully tested practical detection system. | Identified shift from Email Phishing → Smishing, Vish-ing, Quishing with increased attack complexity and need for AI-based detection and user awareness strategies. |
| Marie Weinz, Nicola Zannone, Luca Allodi, Giovanni Apruzzese ,2025, Emerging Trends in Phishing: A Look at Smishing, Vish-ing, Quishing                                                                  | To study how quishing and LLM-generated phishing emails affect organizations, and to check how effective current defenses are against these new phishing threats. | Random Forest, SVM, CNN | It is mainly an evaluation/simulation study, so it does not propose a new standalone detection algorithm or security tool.                                                 | Quishing was found to be as effective as traditional phishing in getting users to the landing page, and LLM-generated emails were also very effective.              |

|                                                                                                                                                                       |                                                                                                                                                                                                                                              |                                                                                                     |                                                                     |                                                                                                                                                                                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pavan Hasti, 2025, SMS Phishing Detection Using Machine Learning and Deep Learning Techniques                                                                         | To construct a New Zealand domain-specific SMS phishing detection system by building effective machine learning (ML) and deep learning (DL) models to accurately detect and categorize "smishing" messages while addressing class imbalance. | Random Forest, XGBoost, CNN, RNN, and LSTM.                                                         | Resource Demands, Contextual Understanding, SMOTE Artificial Noise. | The XG-Boost classifier achieved highest accuracy of 97.05. Random Forest with 96.63 accuracy. LSTM performed with 92.98 accuracy, CNN reached 90.88 and RNN reached 83.71.                                                                                            |
| Ganesh S. Nayak, Balachandra Muniyal, and Manjula C. ,2025, Enhancing Phishing Detection: A Machine Learning Approach With Feature Selection and Deep Learning Models | To improve the accuracy and cost efficiency of phishing detection by analyzing a large dataset of URLs, employing feature selection to reduce the number of required characteristics, and utilizing multiple deep learning classifiers       | Feedforward Neural Network (FNN), Deep Neural Network (DNN), Wide and Deep Model (WDM), and TabNet. | Accuracy-Recall Trade-of, Dataset Evolution                         | The Feedforward (FNN) model achieved a 94.46 accuracy using only 14 selected features. The FNN model stood out with the highest "anti-phishing score" of 0.9521. Performance metrics for other models included DNN (93.92), TabNet (93.82), and Wide and Deep (91.39). |



## RESEARCH GAP AND RESEARCH CHALLENGES

### 5.1 Research Gap

#### 1. Lack of unified testing and monitoring across SMS and QR channels:

Most existing systems focus only on either SMS phishing detection or QR-code phishing detection separately. They do not provide a combined framework that can analyze both attack channels together. Because modern phishing attacks often use SMS messages and QR codes simultaneously, separate monitoring reduces overall detection efficiency.

#### 2. Lack of continuous validation because many systems work only at fixed milestones:

Many existing phishing detection systems perform analysis only at predefined stages or after the user interacts with suspicious content. They do not continuously monitor incoming messages and QR scans in real time. This delay increases the risk of users accessing malicious websites before detection occurs.

#### 3. Lack of contract-based or context-based validation for message and link safety:

Most traditional systems check only keywords, URLs, or blacklisted domains without understanding the relationship between the message content and the linked destination. They fail to verify whether the sender identity, SMS context, and QR-code URL logically match each other. This limitation allows contextual phishing attacks to bypass detection mechanisms.

#### 4. Manual root cause analysis across multiple logs and tools:

Existing approaches often require security analysts to manually inspect logs, alerts, URLs, and user activities using multiple tools. This process consumes time and increases complexity during phishing investigation. Lack of automated analysis also delays threat identification and response.

#### 5. Ineffectiveness in detecting behavioral drift, new domains, and evolving phishing tactics:

Attackers continuously change phishing techniques, domain structures, URL patterns, and social engineering methods to avoid detection. Many existing systems depend heavily on old datasets and blacklist-based approaches, making them weak against zero-day attacks and newly generated malicious domains.

#### 6. Lack of intelligent recommendations, risk scoring, and preventive action in many existing systems:

Several phishing detection systems only classify content as safe or unsafe without providing meaningful guidance to users. They do not generate risk scores, explain suspicious behavior,

or suggest preventive actions such as blocking or ignoring malicious links. As a result, users may still interact with harmful content even after detection.

## **5.2 Research Challenges**

### **1. Designing a continuous monitoring mechanism that does not degrade mobile performance**

The system must continuously monitor SMS messages and QR-code scans in real time without affecting device speed, battery life, or memory usage. Achieving lightweight and efficient background processing is a major challenge in mobile environments.

### **2. Building an effective validation approach using status, urgency, sender behavior, and URL features**

Phishing detection requires analysis of multiple factors such as message tone, urgency indicators, sender identity, suspicious words, and URL characteristics. Combining all these features into a reliable validation framework while maintaining high accuracy is technically challenging.

### **3. Managing large volumes of SMS and QR test results and keeping them organized for analysis**

The system may process a huge number of SMS messages and QR-code scans daily. Properly storing, organizing, and analyzing these results for training, testing, and future improvements requires efficient data management techniques.

### **4. Detecting issues in real time and providing immediate warning before user interaction**

The detection process must be fast enough to alert the user before they click a malicious link or scan an unsafe QR code. Providing accurate real-time predictions with minimal delay is difficult because the system needs to analyze multiple features instantly.

### **5. Computing metrics such as uptime, latency, failure rate, and suspicious pattern frequency efficiently**

The system should continuously measure performance-related metrics to ensure reliability and stability. Calculating these metrics accurately without increasing computational overhead is a significant technical challenge.

### **6. Designing a rule-based or hybrid recommendation engine that can produce meaningful guidance**

The project aims not only to detect phishing attacks but also to guide users with proper recommendations and preventive actions. Designing a hybrid recommendation engine that

combines AI predictions with rule-based logic while remaining understandable to users is challenging.

**7. Ensuring scalability and modularity so that future versions can include more models and more attack types**

The architecture should support future improvements such as additional machine learning models, multilingual support, cloud integration, and detection of new phishing techniques. Building a scalable and modular system without increasing complexity is an important challenge during development.

## PROPOSED METHODOLOGY

The proposed system, Real Time AI/ML Based Quishing and Smishing Detection and Prevention System, is designed to detect phishing attacks delivered through SMS messages and QR codes in real time. The methodology combines Artificial Intelligence (AI), Machine Learning (ML), Natural Language Processing (NLP), QR-code analysis, URL verification, and contextual correlation techniques to identify malicious activities and warn users before interaction.

The architecture of the system is divided into multiple layers, where each layer performs a specific task in the detection and prevention process. The overall workflow starts from data collection and ends with user alerts, prevention actions, and continuous model improvement.

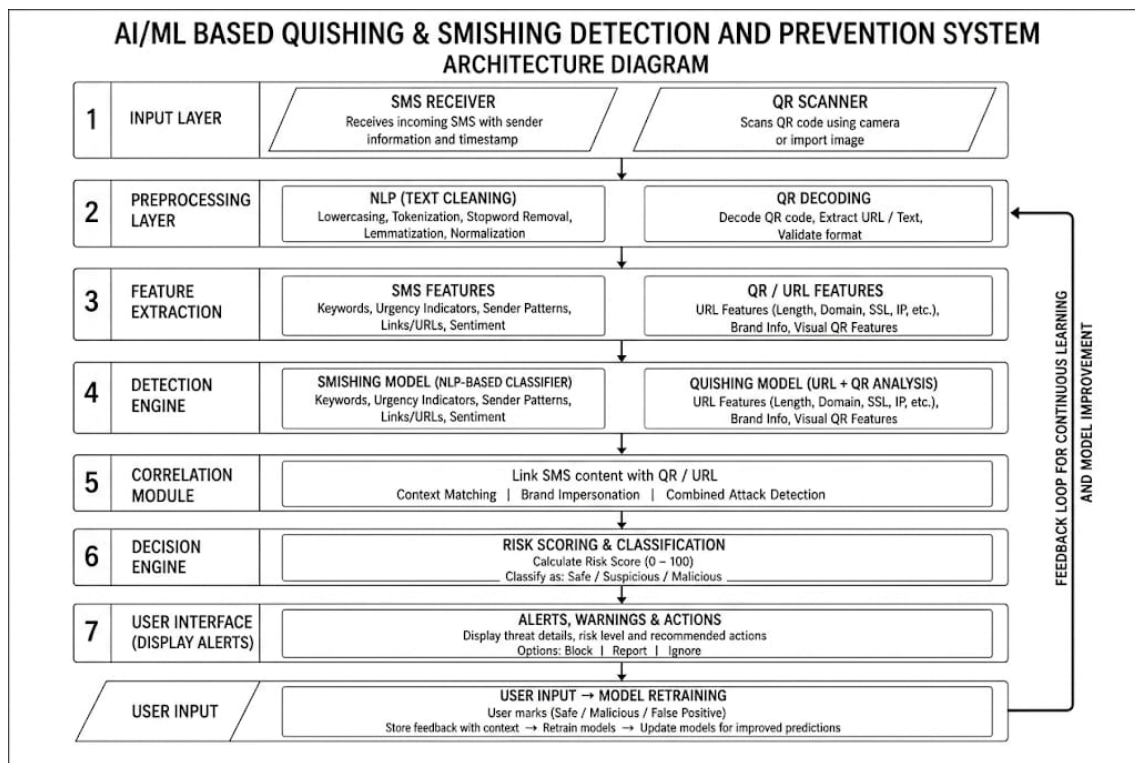


Fig 6.1 Methodology diagram

### 6.1 Input Layer

The input layer is responsible for collecting data from two major sources: SMS messages and QR codes.

The SMS Receiver Module continuously monitors incoming SMS messages and collects information such as message content, sender details, timestamps, and embedded links. This allows the system to inspect suspicious messages in real time. The QR Scanner Module captures QR codes using the mobile camera or imported images. After scanning, the QR code is decoded to extract hidden

URLs, text, or embedded information. This enables the system to analyze whether the QR code leads to safe or malicious destinations.

The input layer acts as the first stage of data acquisition and ensures that all relevant information is forwarded for further processing.

## 6.2 Preprocessing Layer

The preprocessing layer prepares the collected data for analysis by removing unnecessary information and converting the raw input into a structured format.

For SMS messages, Natural Language Processing (NLP) techniques are applied. The preprocessing operations include:

- Lowercasing
- Tokenization
- Stop-word removal
- Lemmatization
- Normalization

These techniques help clean noisy SMS text and improve the accuracy of phishing detection.

For QR codes, the decoding process extracts the hidden URL or text payload from the QR image. The extracted content is validated and normalized before moving to the feature extraction stage.

This layer improves data quality and ensures that the detection engine receives meaningful input.

## 6.3 Feature Extraction Layer

The feature extraction layer identifies important characteristics from SMS messages and QR-code URLs.

### SMS Features

The system extracts several SMS-related features such as:

- Suspicious keywords
- Urgency indicators
- Sender behavior patterns

- Links and embedded URLs
- Sentiment and tone analysis

These features help identify phishing attempts that use fear, urgency, rewards, or fake notifications to manipulate users.

### **QR / URL Features**

For QR-code analysis, the system extracts URL-related features including:

- URL length
- Domain information
- SSL certificate presence
- IP address usage
- Redirect behavior
- Brand impersonation patterns
- Visual QR characteristics

These features help detect malicious URLs hidden inside QR codes. The extracted features are forwarded to the AI/ML detection engine for classification.

## **6.4 Detection engine**

The detection engine is the core component of the system and uses AI/ML models to classify phishing attacks.

### **Smishing Detection Model**

The SMS analysis module uses NLP-based machine learning models to identify phishing messages. The model examines message semantics, suspicious language, urgency, sender identity, and phishing patterns to determine whether the SMS is legitimate or malicious.

### **Quishing Detection Model**

The QR-code analysis module examines decoded URLs and QR-related features. AI/ML algorithms analyze domain structure, suspicious URL behavior, unsafe redirects, and phishing indicators to classify the QR content.

Both models work independently but contribute to the final security decision.

## 6.5 Correlation Module

The correlation module links SMS content with QR-code or URL information to detect contextual phishing attacks.

This module checks whether:

- The SMS message matches the destination URL
- The sender identity corresponds to the linked website
- Brand impersonation is present
- Multiple attack channels are combined

For example, if a message claims to come from a bank but the QR code redirects to an unrelated domain, the system identifies this as suspicious behavior.

The correlation mechanism improves detection accuracy by analyzing relationships between different attack components.

## 6.6 Decision Engine

The decision engine calculates the final risk score based on outputs received from the smishing model, quishing model, and correlation module.

The system generates:

- Risk score (0–100)
- Threat classification
- Security level

The detected content is classified into categories such as:

- Safe
- Suspicious
- Malicious

This classification helps users understand the severity of the threat clearly and quickly.

## 6.7 User Interface and Alert System

The user interface displays the detection results and provides security recommendations. When suspicious activity is detected, the system generates:

- Warning messages
- Risk-level indicators
- Security alerts
- Preventive suggestions

The application may provide actions such as:

- Block
- Report
- Ignore

This layer ensures that users receive immediate guidance before interacting with harmful links or QR codes.

## 6.8 Feedback Module

The feedback module allows users to provide feedback on detection results. Users can mark messages or QR codes as:

- Safe
- Malicious
- False positive

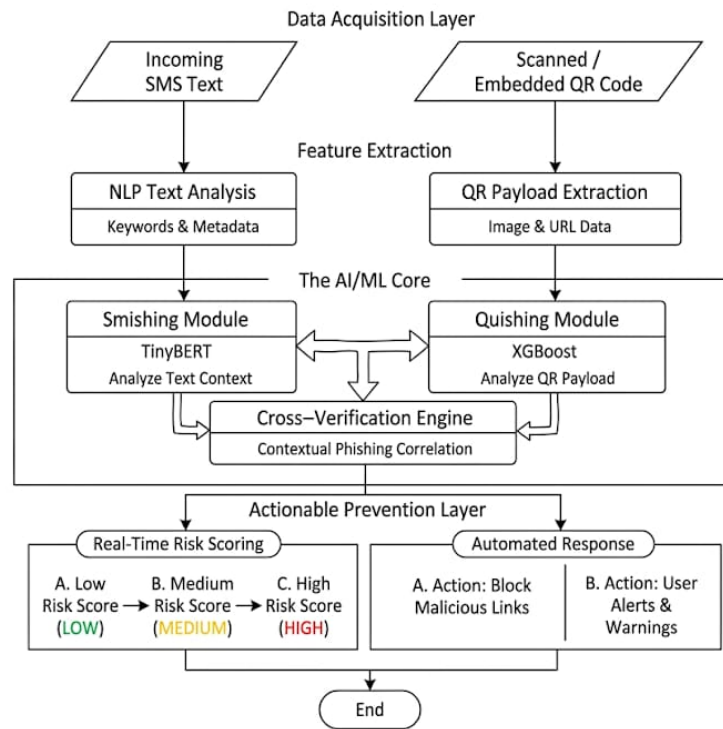
The feedback data is stored and used for model retraining and continuous improvement. This mechanism helps the system adapt to new phishing techniques and improve detection accuracy over time.

The feedback loop also supports continuous learning, making the system more effective against evolving phishing attacks.



## SYSTEM ARCHITECTURE

The system architecture for the Real-Time AI/ML Quishing and Smishing Detection and Prevention System is designed as a modular, multi-tier framework. It enables the simultaneous processing of text-based and image-based phishing vectors to provide a unified security solution.



**Fig 7.1** System Architecture Diagram

### 7.1 Data Acquisition Layer

This layer serves as the primary interface between the mobile environment and the detection engine. It is responsible for gathering raw data from two distinct communication streams.

- **SMS Monitoring Module:** Utilizing the Android SMS API, the system continuously monitors incoming messages in real-time. It captures the sender's information, the message body, and any embedded hyperlinks.
- **QR Code Scanning Module:** This module uses the ZXing (Zebra Crossing) library to scan and decode QR codes from various sources, such as physical stickers, digital displays, or images within the gallery.
- **Real-Time Processing:** Both streams operate concurrently, ensuring that threats are intercepted before the user interacts with malicious content.

## 7.2 Feature Extraction and Preprocessing Layer

Before the data is passed to the AI models, it undergoes rigorous transformation to convert raw inputs into structured features suitable for machine learning analysis.

- **NLP Text Preprocessing:** The system employs the SpaCy library for Natural Language Processing (NLP). This involves tokenization, lemmatization, and stop-word removal to clean the SMS text.
- **Contextual Feature Engineering:** Key features such as "urgency keywords" (e.g., "Immediate Action Required," "Account Blocked"), sender reputation, and the presence of suspicious short-URLs are extracted.
- **QR Payload Analysis:** Once a QR code is decoded, the system extracts the payload, which is typically a URL. The system also analyzes the visual properties of the QR code to detect potential tampering or "over-sticking" (Attacker placing a fake QR over a legitimate one).

## 7.3 The AI/ML Core (Hybrid Detection Pipeline)

This is the "brain" of the system, where sophisticated models analyze the extracted features to identify zero-day threats that traditional blacklists might miss.

### 1. Smishing Module (TinyBERT)

- **Model Selection:** The system uses TinyBERT, a lightweight version of the BERT (Bidirectional Encoder Representations from Transformers) model, optimized for mobile deployment.
- **Functionality:** It analyzes the semantic meaning of the SMS content. Unlike keyword-based filters, TinyBERT understands the context of a message, allowing it to distinguish between a legitimate bank alert and a deceptive phishing lure.

### 2. Quishing Module (XGBoost)

- **Model Selection:** For URL safety classification, the XGBoost (Extreme Gradient Boosting) algorithm is utilized due to its high accuracy and speed.
- **Functionality:** It evaluates the safety of the URL extracted from the QR code by analyzing domain age, SSL certificate status, and URL structure (e.g., typosquatting like g00gle.com instead of google.com).

### 3. Cross-Verification Engine

- **Contextual Correlation:** This unique engine bridges the gap between the two modules. It correlates the intent of an SMS with the landing page of a QR code.
- **Example Case:** If a user receives an SMS regarding "Electricity Bill Payment" but the scanned QR code leads to a "Social Media Login" page, the system flags this as a highly suspicious contextual phishing attack.

## 7.4 Actionable Prevention Layer

The final layer focuses on user safety by providing immediate feedback and automated safeguards.

- **Real-Time Risk Scoring:** The system calculates a unified risk score based on the outputs of the AI modules. Threats are categorized into three levels:
  - **Low (Green):** Safe to proceed.
  - **Medium (Yellow):** Potential risk; user is advised to be cautious.
  - **High (Red):** Confirmed threat; access is restricted.
- **Automated Blocking and Alerts:** If a threat is rated as "High," the system automatically blocks the malicious link and triggers a full-screen security warning. It provides the user with visual safety indicators and specific reasons why the content was flagged.

## 7.5 Technical Implementation Details

To ensure the system is feasible and operational within a mobile environment, the following stack is employed:

- **Mobile Frontend:** Mobile Frontend: Developed using Android Studio with Kotlin/Java, providing a responsive dashboard for the user to view security logs and real-time alerts.
- **Backend Integration:** A FastAPI server (Python-based) acts as the bridge between the mobile app and the AI/ML models, ensuring fast response times and low latency.
- **Resource Management:** By using TinyBERT and optimized ML models, the system minimizes computational overhead, ensuring it does not drain the device battery or slow down performance

## EXPECTED OUTCOMES

The proposed Real Time AI/ML Based Quishing and Smishing Detection and Prevention System is expected to provide an intelligent, secure, and efficient solution for detecting phishing attacks delivered through SMS messages and QR codes. The system aims to improve mobile security by identifying suspicious activities in real time and preventing users from interacting with malicious links or fake websites.

### 1. Real-Time Phishing Detection

The system is expected to continuously monitor incoming SMS messages and scanned QR codes in real time. It will identify suspicious activities instantly and provide alerts before the user interacts with malicious content.

### 2. Accurate AI/ML-Based Classification

The project aims to achieve high phishing detection accuracy using Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP) techniques. The models will classify SMS messages and QR-code URLs as safe, suspicious, or malicious.

### 3. Unified Smishing and Quishing Detection

The proposed system will provide a single integrated framework capable of detecting both SMS phishing (smishing) and QR-code phishing (quishing). This improves security compared to systems that focus on only one attack type.

### 4. Detection of Malicious URLs and Fake Domains

The system is expected to analyze URLs, domains, redirects, and suspicious website patterns to identify phishing links hidden inside SMS messages and QR codes.

### 5. Context-Aware Verification

The project will compare SMS content with embedded links or QR-code destinations to verify whether the sender and destination are logically related. This helps detect fake messages and brand impersonation attacks.

### 6. Risk Scoring and Threat Classification

The system will generate risk scores and classify detected content into categories such as safe, suspicious, or malicious. This helps users understand the severity of threats easily.

### 7. Real-Time Alerts and Preventive Actions

The application will provide warning notifications, security alerts, and preventive recommendations before the user opens suspicious links or scans malicious QR codes.

#### **8. Reduced False Positives**

By combining AI/ML models with contextual verification, the system is expected to reduce incorrect phishing predictions and improve overall detection reliability.

#### **9. Lightweight Mobile-Friendly Performance**

The system is expected to operate efficiently on Android devices without causing significant battery drain, memory consumption, or performance degradation.

#### **10. Continuous Learning and Improvement**

The feedback mechanism will help the system learn from new phishing patterns and user responses. This supports continuous model improvement and adaptation to evolving cyber threats.

#### **11. Improved User Awareness**

The project will educate users about phishing risks by displaying clear alerts, threat details, and security recommendations during suspicious activities.

#### **12. Enhanced Mobile Security**

The proposed system is expected to improve overall mobile cybersecurity by protecting users from smishing, quishing, malicious URLs, and emerging phishing attacks.

#### **13. Scalable and Extensible Architecture**

The architecture will support future enhancements such as additional AI models, multilingual support, cloud integration, and detection of more advanced cyber threats.

## **SOCIETAL IMPACT, STANDARDS AND SUSTAINABILITY**

### **9.1 Societal Impact**

The proposed Real Time AI/ML Based Quishing and Smishing Detection and Prevention System has significant societal impact by improving digital security and protecting users from modern cyber threats. As smartphones and digital payment systems are widely used in daily life, phishing attacks through SMS messages and QR codes have become increasingly common. The proposed system helps reduce financial fraud, identity theft, and unauthorized access to personal information by detecting malicious links and suspicious activities in real time.

The system improves public awareness about cybersecurity threats such as smishing and quishing by providing alerts, warnings, and preventive recommendations. It helps users make safer decisions while interacting with SMS messages, online payment links, and QR codes. This contributes to safer digital communication and encourages secure usage of online services.

The project also supports organizations, banks, educational institutions, and digital payment platforms by reducing phishing-related risks and improving trust in mobile transactions. By preventing cyber fraud and enhancing digital safety, the system contributes positively to society and supports the development of secure digital ecosystems.

### **9.2 Standards**

The proposed system follows standard cybersecurity and software engineering practices to ensure reliability, security, and efficiency.

The system follows secure coding practices for Android application development and AI/ML implementation. Standard Machine Learning and Natural Language Processing techniques are used for phishing detection and classification.

URL validation and QR-code analysis are performed using standard security verification methods. The project follows Android security permissions and privacy guidelines while accessing SMS messages and camera functionality. Data storage and user information handling are performed securely to maintain privacy and confidentiality. The system architecture is designed to be modular, scalable, and maintainable according to modern software development standards.

These standards help improve system performance, usability, and cybersecurity reliability.

### 9.3 Sustainability

The proposed system is designed with sustainability in terms of scalability, adaptability, and efficient resource utilization.

The system uses lightweight AI/ML models to minimize battery consumption, memory usage, and processing overhead on mobile devices. Continuous learning mechanisms allow the system to adapt to new phishing patterns without requiring complete redevelopment. The modular architecture supports future upgrades, additional security features, and integration of advanced AI technologies. The project promotes long-term digital safety by continuously protecting users against evolving phishing attacks. By preventing cyber fraud and financial losses, the system contributes to a safer and more sustainable digital environment.

The proposed framework ensures technical sustainability, future scalability, and long-term cybersecurity support for mobile users.

## **CONCLUSION**

The proposed Real Time AI/ML Based Quishing and Smishing Detection and Prevention System provides an intelligent and efficient solution for protecting users against modern phishing attacks delivered through SMS messages and QR codes. With the rapid growth of mobile banking, digital payments, online transactions, and QR-based services, phishing attacks such as smishing and quishing have become major cybersecurity threats. Traditional security mechanisms are often unable to detect these evolving attacks effectively, especially in real-time mobile environments.

The proposed system integrates Artificial Intelligence (AI), Machine Learning (ML), Natural Language Processing (NLP), QR-code analysis, URL verification, and contextual correlation techniques to identify malicious activities accurately. The system continuously monitors incoming SMS messages and scanned QR codes, extracts phishing-related features, and performs intelligent classification to detect suspicious behavior. By combining SMS analysis and QR-code analysis into a unified framework, the system improves phishing detection accuracy and provides better protection against coordinated attacks.

The project also introduces context-aware verification, risk scoring, real-time alerts, and preventive recommendations to help users avoid interacting with malicious links and fake websites. The feedback mechanism supports continuous learning and adaptation, enabling the system to handle evolving phishing tactics and zero-day attacks more effectively.

Overall, the proposed system enhances mobile security, improves user awareness, reduces phishing risks, and provides a scalable and extensible cybersecurity framework for future developments. The project demonstrates how AI/ML technologies can be effectively utilized to build smart and real-time phishing detection systems capable of addressing modern cyber threats in mobile environments.



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